

VayuCoupler App — Download & Setup Guide

Air Pollution–Weather Coupled Forecasting & Predictive GRAP System (Delhi NCR)

1. System Overview & The Core Innovation

VayuCoupler is a disaster-management decision support system developed for the Ministry of Earth Sciences. Unlike standard reactive GRAP (which triggers curbs after smog strikes), VayuCoupler dynamically couples **Planetary Boundary Layer Height (PBLH) compression**, **Thermal Inversion (ΔT) trapping**, and **NASA FIRMS satellite stubble fires** to forecast severe AQI spikes **24 to 72 hours in advance**.

2. Quick Start: Running on Laptop / Desktop (Mac / Windows / Linux)

Follow these 2 simple steps to launch the complete Command Center & API:

Step 1: Open Terminal / Command Prompt

Navigate to the project folder on your computer:

```
cd sih-coupled-aqi-delhi
```

Step 2: Run the One-Click Startup Script

This automatically initializes the environment and launches the server:

```
./run.sh      (or 'python3 run.py' on Windows)
```

Step 3: Open in Browser

Go to ■ <http://127.0.0.1:8000> in Chrome, Safari, or Edge.

3. How to Install as Mobile App on iPhone & Android

The application has a mobile-optimized PWA layout with touch scrubber and bottom navigation bar:

■ For Apple iPhone / iPad (iOS)

1. Connect your iPhone to the same Wi-Fi as your laptop.
2. Open **Safari** and enter: `http://<your-laptop-ip>:8000`
3. Tap the **Share Button** (square with arrow up) at the bottom.
4. Scroll down and tap **'Add to Home Screen'** (■).
5. Tap **'Add'** in the top-right corner.

Now the app opens full-screen like a native iOS app!

■ For Android Phones (Google Chrome)

1. Connect your Android phone to the same Wi-Fi.
2. Open **Chrome** and visit: `http://<your-laptop-ip>:8000`
3. Tap the **Three-Dot Menu** (■) in top-right.
4. Tap **'Install app'** or **'Add to Home screen'**.
5. Tap **'Install'** on the confirmation popup.

The VayuCoupler app icon will appear on your home launcher!

4. Key App Features & Navigation Guide

Tab / Feature	What it Does & Key Talking Point for Judges
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Command Center	Real-time Delhi NCR map with 16 stations, animated wind particle vector field, upwind stubble fire clusters, and 72-hour forecast confidence curves.
Predictive GRAP	Evaluates 9 configurable rules; shows exact lead-time gained (+72h, +48h, +36h, +24h) compared to reactive GRAP.
VayuAI Copilot	Conversational GenAI decision assistant answering questions on stubble trajectories, traffic police orders, and asthma health risks.
Clean Air Windows	24-hour diurnal outdoor planner identifying safe clean windows (1 PM–4 PM) vs dangerous nocturnal inversion trapping (6 AM–9:30 AM).
Atmospheric 3D	Graphic atmospheric cross-section showing boundary layer collapse (1400m to 240m) and thermal inversion capping.
Report Fire (AI)	Citizen crowdsourcing tool with mock Vision AI (94% confidence) to geotag farm fires and auto-dispatch municipal patrol teams.
What-If Simulator	Interactive sliders to test policy interventions (50% stubble reduction, 40% truck bypass) showing instant avoided AQI points.

5. 30-Second Winning Pitch Script for Hackathon

*"Respected Judges, Delhi's current GRAP is reactive — curbs are imposed only after severe smog is confirmed. **VayuCoupler** couples dynamic atmospheric physics (boundary layer collapse, thermal inversion) with satellite fire counts to predict spikes **48 to 72 hours in advance**. This buys 2 to 3 days of lead time for Punjab agriculture departments to accelerate bio-decomposer spraying, traffic police to divert 60,000+ diesel trucks to bypass expressways, and schools to transition online without morning panic."*